

Document Trap

Your season-end basketball tournament starts next week, so you want to brush up on the rules. You upload your league's 100-page rulebook and ask: "How many fouls until I'm out of the game?"

The AI answers: "Five fouls and you foul out." But page 47 says that in tournament games, players foul out on the sixth foul. The AI pulled the standard limit and missed the exception. The answer wasn't made up. It was incomplete.

You can drag in a PDF, paste a Google Doc, or upload a worksheet, and the AI will answer questions about it. It's one of the most common ways you'll use AI for school, and one of the easiest places to get a confident wrong answer. **The trap is thinking 'uploaded' means 'fully read.'**

WHAT'S ACTUALLY HAPPENING

It would be nice if AI worked like this: you upload a document, the model reads the whole thing top to bottom, then answers your question with the whole document in mind.

That's not always what happens. Sometimes the whole document fits in the context window. Often it doesn't. And even when a lot fits, the system may search, summarize, or prioritize parts of the file rather than read it cover to cover.

There's a name for what's often happening: **retrieval**. When the system can't load everything, it tries to retrieve the pieces that match your question. Done well, this lets you find a specific answer in a 50-page document in seconds. Done poorly, the wrong pieces get pulled, and the model answers from incomplete evidence.

Back to the rulebook. The system can't always load the whole thing, so it searches for the pieces that match.

How Chunks Get Picked

1 Your document gets split into chunks

Before you ever ask a question, the rulebook gets cut into smaller pieces, each a paragraph or two long. A few hundred chunks total. The chunks sit in a database, waiting.

2 Each chunk gets a meaning vector

Same idea you saw in the Embeddings lesson. Every chunk gets translated into a meaning profile, a list of numbers that captures what the chunk is about. The chunk about technical fouls ends up close in meaning to other rulebook chunks about penalties, referee calls, and player conduct.

3 Your question becomes a vector too. The closest chunks get loaded.

Now you ask: "Can a player call a timeout during a free throw?" Your question gets its own meaning vector. The system finds the chunks whose vectors are closest. Those chunks (not the whole rulebook) get loaded. The model answers based on what got retrieved.

WHAT GOT LOADED

The model never saw the whole rulebook.

It answered from only the chunks that matched your question. If the chunk that mattered didn't get pulled, the answer still sounds confident. It is just missing what it never loaded.

You can't always tell what got retrieved. But you can steer it. Five moves help.

The five moves

1 Point to the section by name

Include keywords from the document in your question. The system uses them to find matching chunks.

VAGUE

What are the rules on flagrant fouls?

POINTED

What does the technical fouls chapter say about flagrant fouls?

2 Ask one question at a time

Multi-part questions force the system to retrieve different chunks for different parts. Break it up. One focused question per turn gives retrieval the best shot.

COMBO QUESTION

Summarize the policy, list the exceptions, and tell me what to do.

ONE THING AT A TIME

What does the policy say about late assignments? Quote the line.

3 Paste the section directly into your prompt

If you already know the paragraphs that matter, paste them into your message. That puts them in context with no retrieval lottery.

RETRIEVAL LOTTERY

Does my technical fouls clause apply if both players are involved?

PASTED IN DIRECTLY

Here's the clause: 'A technical foul is assessed when...' Does this apply if both players are involved?

4 Upload only what's relevant

Instead of the whole rulebook, upload just the chapter you need. Fewer chunks competing for retrieval slots means the right ones are more likely to make the cut.

WHOLE RULEBOOK

[Uploads the entire 100-page rulebook] What does it say about flagrant fouls?

JUST THE CHAPTER

[Uploads only the 3-page fouls chapter] What does it say about flagrant fouls?

5 Ask the AI to quote

Add 'quote the exact section you're basing this on' to your question. If the quote is missing or doesn't match the document, retrieval probably failed.

NO RECEIPTS

What does the rulebook say about flagrant fouls?

ASK FOR THE QUOTE

What does the rulebook say about flagrant fouls? Quote the exact line you're basing it on.

All five share one idea: make the right chunks easy to find. The more specific your question, the more reliable the answer.



Uploaded does not mean understood. Confident does not mean complete. Asking for evidence is the skill.

It answers from what it retrieved.

Not from having read the whole thing.